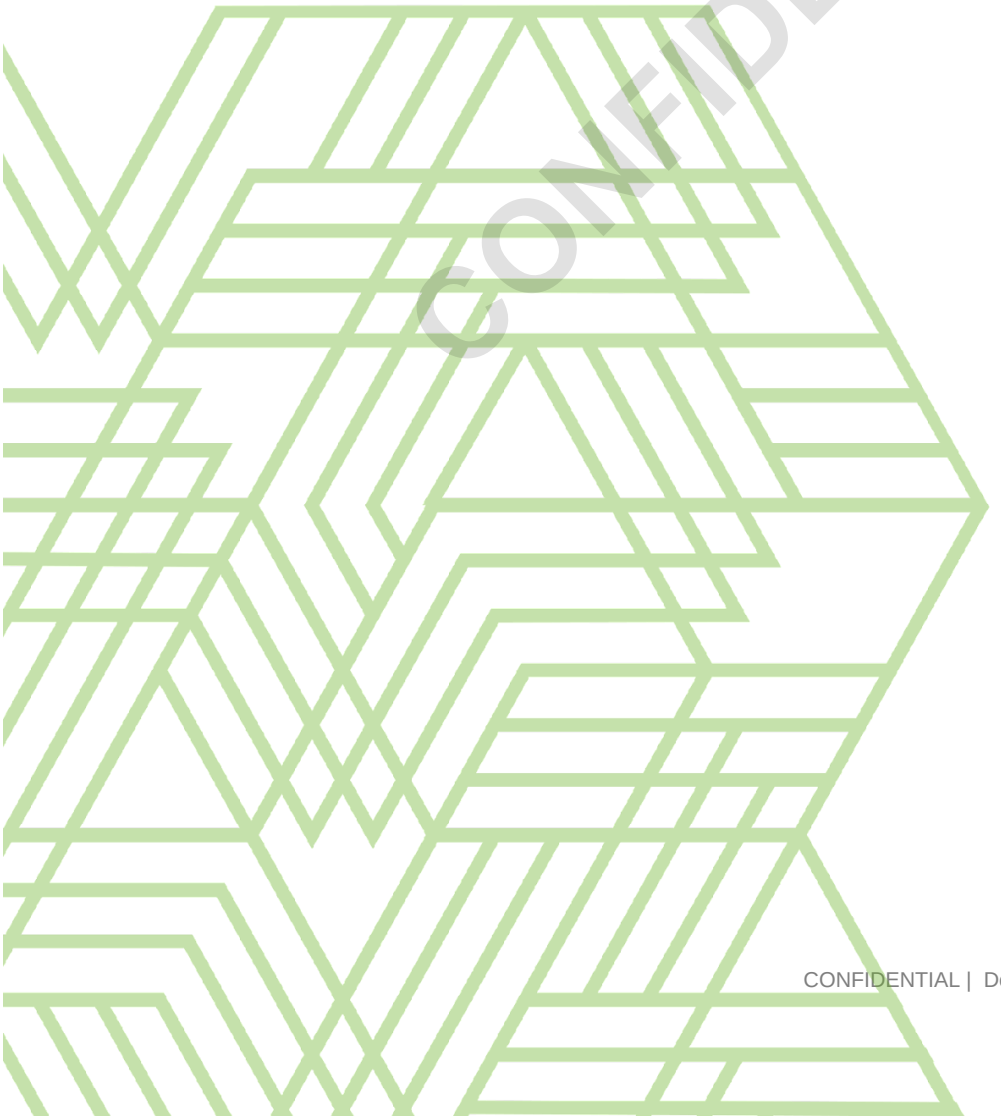
ATM2/ATM3/ATM33 Wireless SoC Family Keil MDK

Configuration Guide

SUMMARY: This document provides the installation and configuration required to use Keil MDK for Atmosic SDK on Windows OS.

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1. Overview

This document provides the installation and configuration required to use Keil MDK for Atmosic SDK on Windows OS.

1.1 Prerequisite

1) Atmosic SDK 5.1 or later
2) ATM33 platform
    a) ATM3330e/ATM3330 Evaluation Board (EVB)
3) ATM2/ATM3 platform
    a) Atmosic ATM/ATM3 Evaluation Board and Interface Board v3.x
    b) J-Link Plus device (SWD Interface) with pin override support
    (Pin override is required for chip reset. Some J-Link models e.g. J-Trace don't support this feature and would get an error during target init process.)
4) J-Link Software Pack (https://www.segger.com/downloads/jlink/JLink_Windows.exe)
5) Keil MDK-ARM (https://www.keil.com/demo/eval/arm.htm)
6) Toolchain
    a) GNU ARM Embedded Toolchain 10.3-2021.07, built-in in Atmosic SDK Windows installer (https://developer.arm.com/tools-and-software/open-source-software/developer-tools/gnu-toolchain/gnu-rm/downloads)
    b) ARM Compiler 6.16, built-in in MDK-ARM v5.34, MDK-ARM license is required (https://developer.arm.com/tools-and-software/embedded/arm-compiler/downloads/version-6)

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1.2 Limitation

Each example in the SDK has two Keil projects named 'keil_gcc' and 'keil_arm' except NVDS-only, bootloader and MCUBoot examples. These Keil projects has few limitations as below:

- Limited SDK functionality
  The Keil projects support the build and program bootloader, example firmware and NVDS data into ATM2/ATM3 and ATM33 devices via the J-Link interface. For the other SDK makefile functions (such as program OTP, pull firmware/NVDS/OTP, show NVDS/OTP data, etc.), please use MSYS2 console in the SDK. Changing J-Link driver for openocd use might be required for the

functions needed to communicate with the device. Refer to 3.6 Using J-Link Debug Probes with openocd.

- NVDS Size Limitation in ATM2/ATM3
  Due to the 4K-align padding in ATM2/ATM3 NVDS bin file generation, the size of NVDS data is limited at 4 Kbyte. The NVDS data which exceed 4 bytes would be truncated in Keil projects.

- Not Support ATM33 OTA
  Since the IDE project generator doesn't support ATM33 OTA required make options USE_MCUBOOT=1 and ATMSTK=1, ATM33 OTA image cannot be built by IDE projects.

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2. ATM2/ATM3 Platform

2.1 Installation

1) Download the Atmosic SDK and follow the REAMDE to setup Universal Serial Bus device (can ignore this if using Atmosic SDK Windows Installer)
   a) Libusb-1.0.dll
   b) WinUSB
   c) Viewer for Console Output
   d) Connect mini-USB cable to [USB1] of Interface Board. Confirm "Atmosic RDI USB1" is showing in Windows device manager under USB devices
   e) Connect mini-USB cable to [USB0] of Interface Board. Confirm "Atmosic RDI USB0" is showing in Windows device manager under USB devices
2) J-Link Debug Probes
   a) Insert the JLink device and Install the JLink Tool Kit
   b) Device manager: After successful installation, "J-Link driver" should show up under Universal Serial Bus controllers.



Figure 1 - J-Link Driver Installation Successful

c) Pin Connection:

- Atmosic Interface Board V2.x:

Remove JP18, JP19 and JP22 jumpers on the interface board.

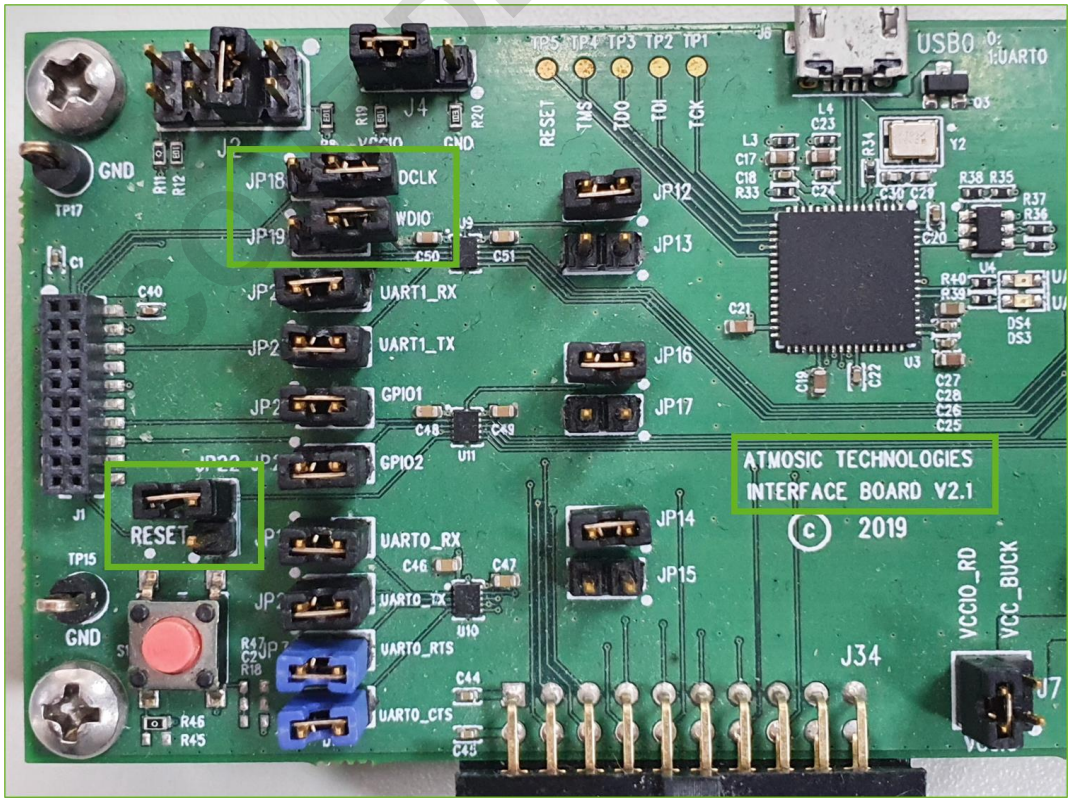


Figure 2 - Interface Board V2.x

Follow Table 1 for Fly-wired connection.

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Table 1 - Fly-wired Connection

Pin	Interface Board	EVB
SWDCLK	JP18	N/A
SWDIO	JP19	N/A
GND	N/A	J4 Pin1
VREF	N/A	J4 Pin2
TDI	N/A	J4 Pin47
RTCK	N/A	J4 Pin49

- Atmosic Interface Board V3.x:

Remove JP18, JP19 and JP22 jumpers on the interface board.
Connect J-Link Adapter to J-Link socket of interface board

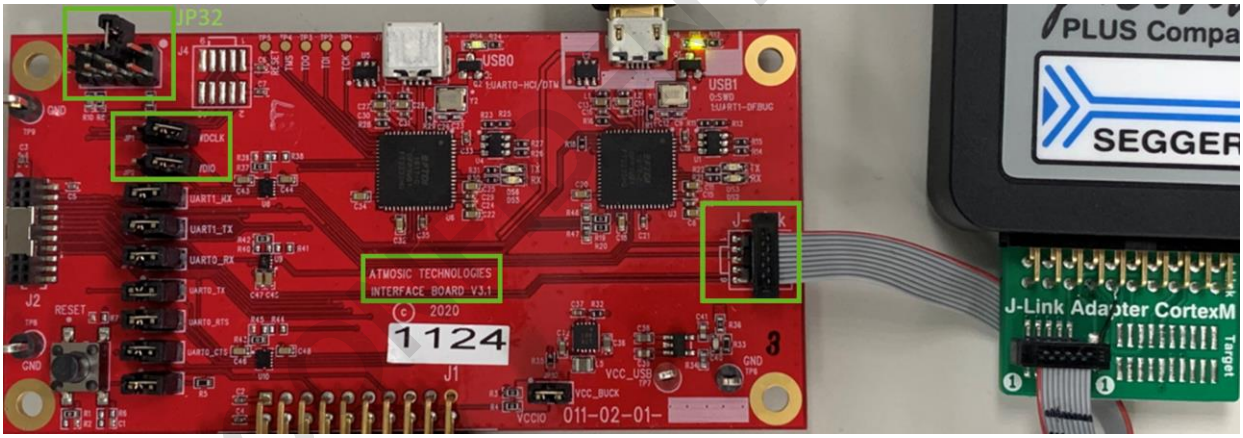


Figure 3 - Interface Board V3.x

3) Install ARM Keil MDK-Lite
4) Install GNU Tools ARM Embedded (can ignore this if using Atmosic SDK Windows Installer)

Add the toolchain path to the user or system path

5) Flash Loader(can ignore this if using Atmosic SDK Windows Installer)
    a) Copy JLinkDevices.xml from {ATM_SDK}\tools\keil to {JLINK_PATH}
    The {JLINK_PATH} depends on the installed J-Link version in the user's environment. It could be "C:\Program Files (x86)\SEGGER\JLink", "C:\Program Files\SEGGER\JLink" or others.
    b) Copy atmx_flash.flm and atmx_nvds.flm from {ATM_SDK}\tools\keil to {JLINK_PATH}\Devices\Atmosic\atmx

c) Copy atmx_flash.FLM and atmx_nvds.FLM from {ATM_SDK}\tools\keil] to C:\Keil_v5\ARM\Flash

6) Open Keil project file with GNU toolchain
   a) GNU toolchain
      i) Open {example_name}.uvprojx in {ATM_SDK}\platform\atm[2,3]\ATM2xxx-xxx\examples\{example_name}\[keil_gcc,keil_arm]

Note: J-Link debug probe will not provide power to the EVB and still need to plug USB cable into USB1 of the EVK to supply the power to the EVB.

2.3 Build Example and Program

1) Program Flash NVDS data for each example(refer to section Flash NVDS Writer)
2) Example project (under keil_gcc/keil_arm folder)
3) Edit/Compile/Program/Debug Code at Keil IDE, see Figure 4 and Figure 5.

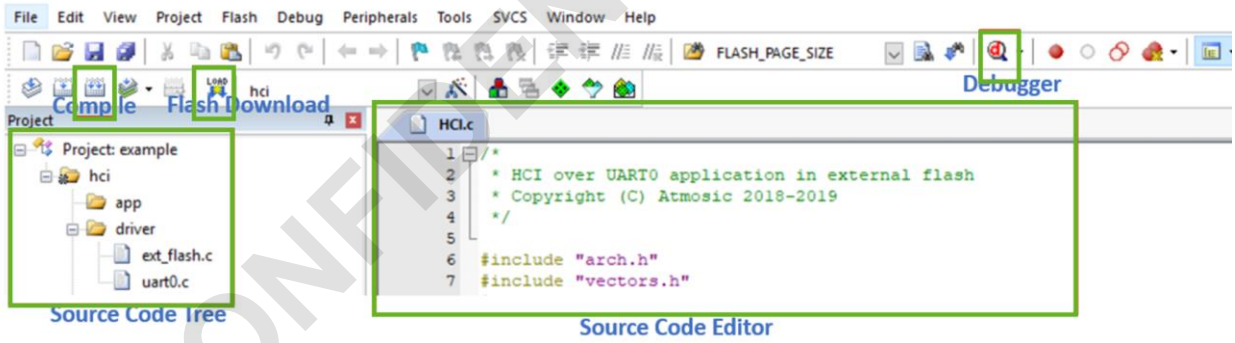


Figure 4 - Keil MDK IDE GUI

4) Run time debugger

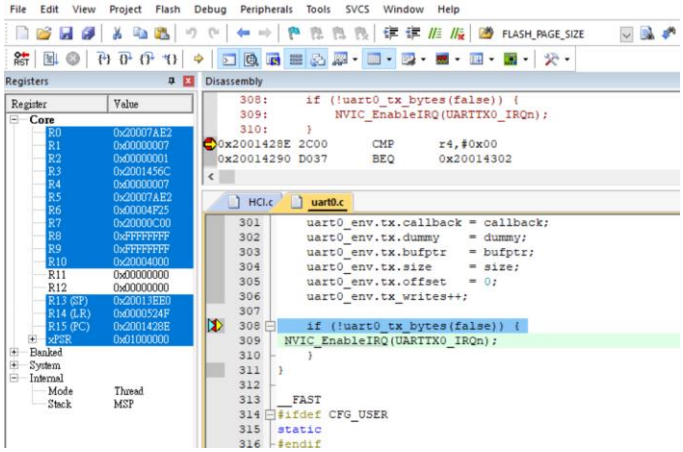


Figure 5 - Debug Session

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2.3.1 Use GCC Compiler in Keil uVersion: (Only for GCC)

When creating a new keil project, follow the steps below to select GNU GCC compiler:

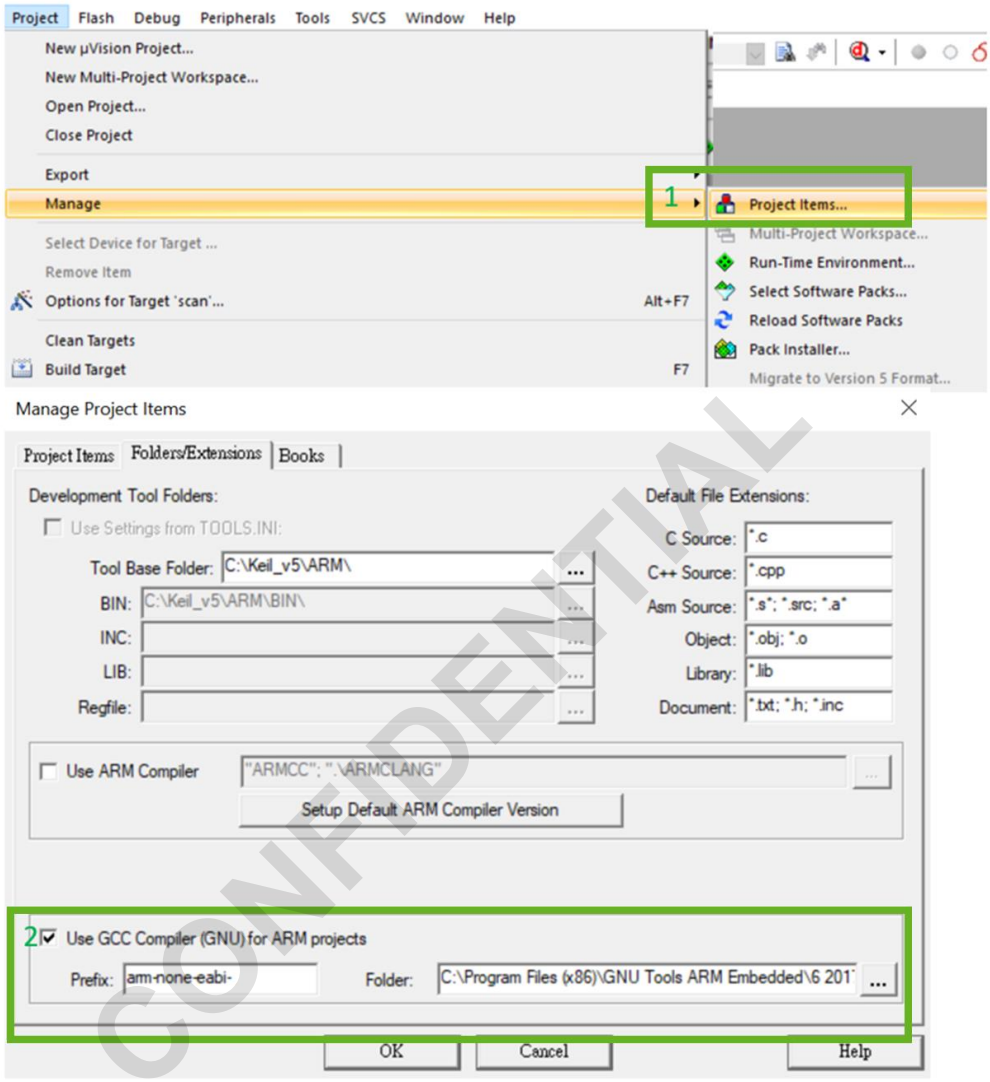


Figure 6 - Select Compiler

2.3.1 Add Flash Programming Algorithm in Keil uVersion

When creating a new keil project, follow the steps below to select a flash programming algorithm.

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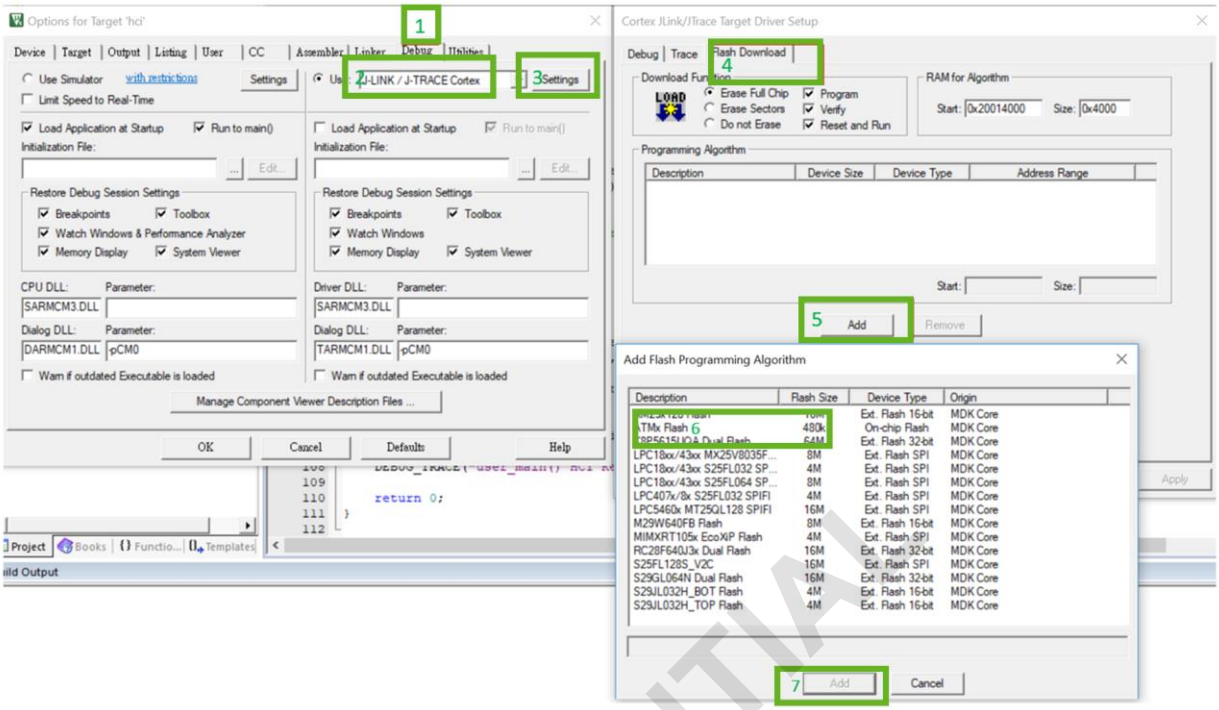


Figure 7 - Add Flash Loader

2.4 Release Build

1) Open the “Option of Target” dialog of Keil IDE and select CC tab page.
2) Remove “CFB_DBG” of “Preprocessor Symbols Define”
3) Modify compile option (Misc Controls): remove “-g3” and use “-flto”
4) Open the “Option for Target” dialog of Keil IDE and select Linker tag page. Add compile option (Misc Controls): “-flto”

2.5 Flash NVDS Writer

The corresponding NVDS data (flash_nvds.bin) would be built with the KEIL project generating make command (keil_gcc_gen/keil_arm_gen) and applying the NVDS related make options to the NVDS bin file. To program NVDS data, click “Flash NVDS Writer” in the “Tools” menu or execute atmosic_nvds.bat batch file in the Keil project folder.

This batch file is added into keil’s customize tools menu in the SDK installation as Figure 8.

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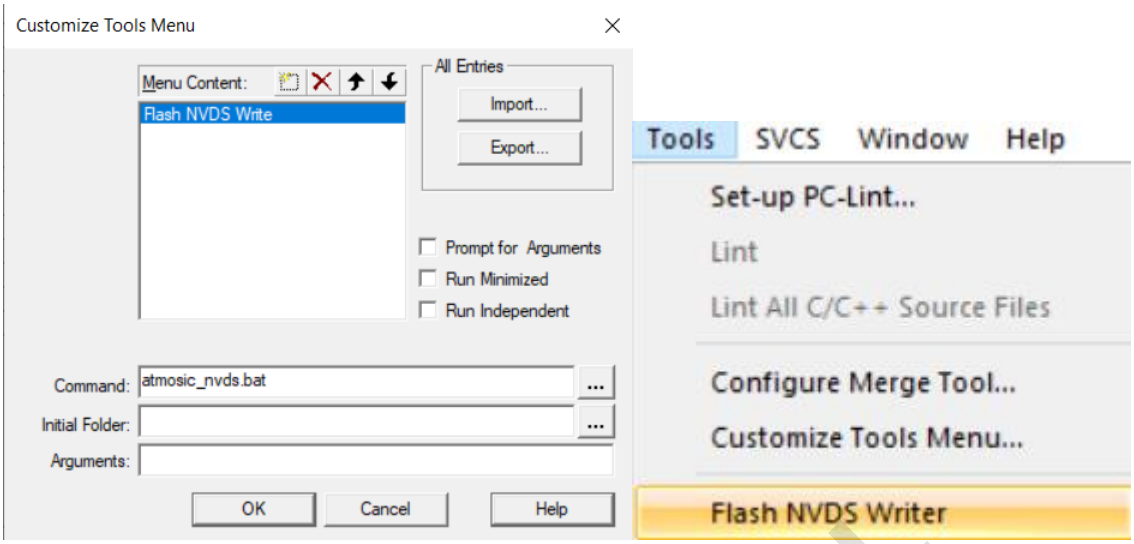


Figure 8 - Customize Tools Menu

2.6 Using J-Link Debug Probes with Openocd

Atmosic SDK tool chain consists of many makefile targets in MSYS2 to perform the functionalities for firmware development. It can be used on J-Link Debug Probes just as the Atmosic Interface Board's SWD interface. We have to uninstall original J-Link driver first and then execute Start → AtmosicSDK → Install RDI interface.

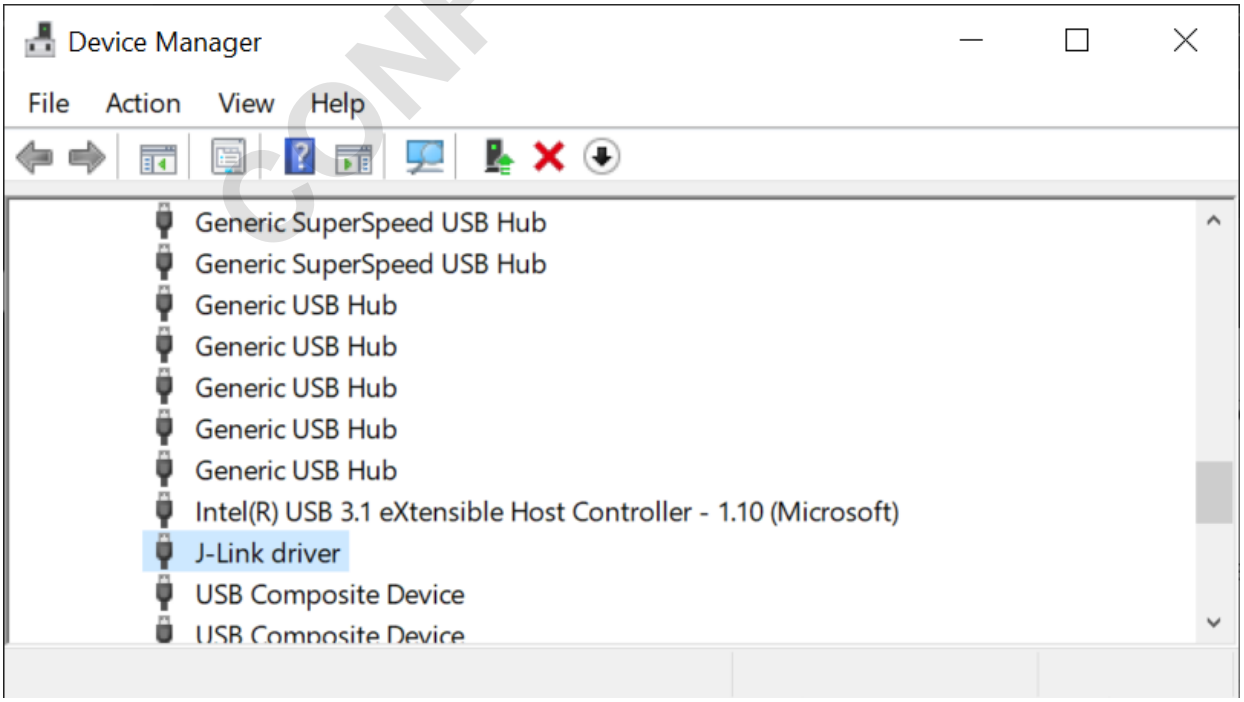


Figure 9 - Uninstall J-Link driver

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After successful installation, there would be a “BULK interface” in “Universal Serial Bus devices”

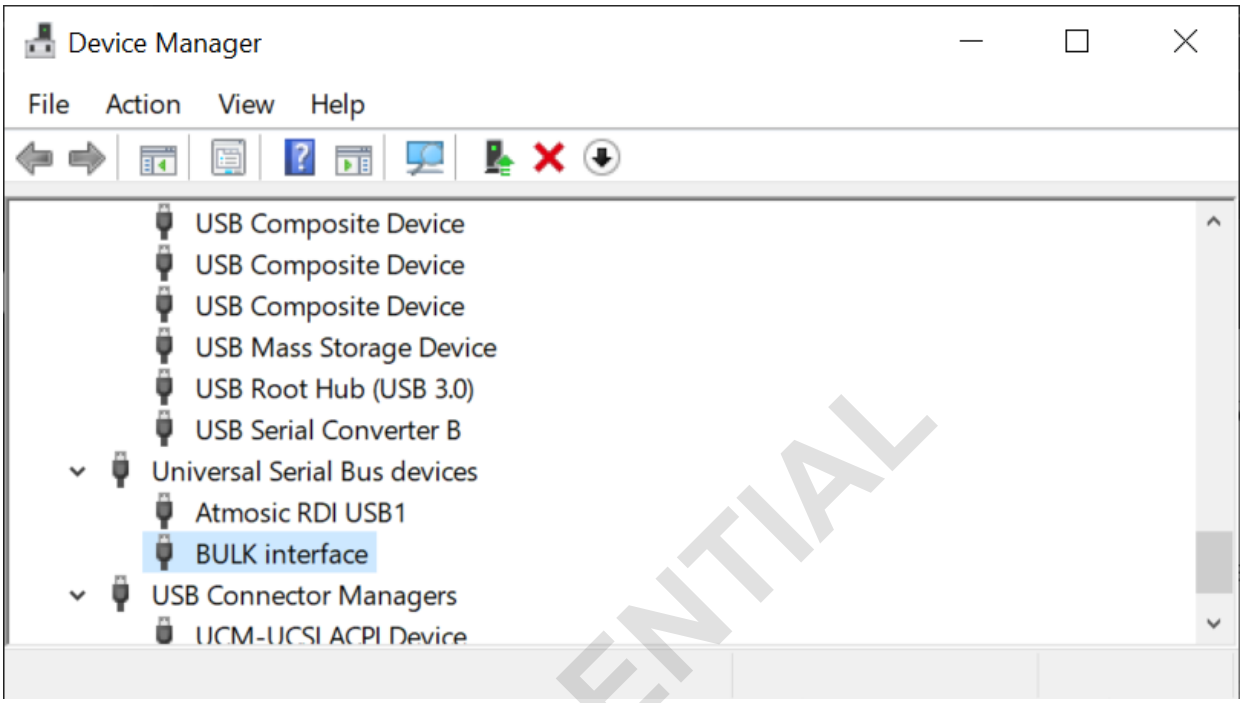


Figure 10 - BULK interface driver for Openocd use in MSYS2

The usage of Makefile helper targets are the same, just appends “SWDIF=JLINK” for each command. (for example: make run_all SWDIF=JLINK). “SWDIF=JLINK” is not required for ATM33 devices since Openocd uses J-Link to access ATM33 by default.

3. ATM33 Platform

3.1 Installation

1) Install Atmosic SDK and follow the REAMDE.Windows in SDK tools folder to setup Universal Serial Bus (USB) device
    a) Connect mini-USB cable to the Micro-B USB Connector on the ATM3330e/ATM3330 Evaluation Board, see Figure 11. Confirm "JLink CDC UART Port" and “J-Link driver” are showing in Windows Device Manager.

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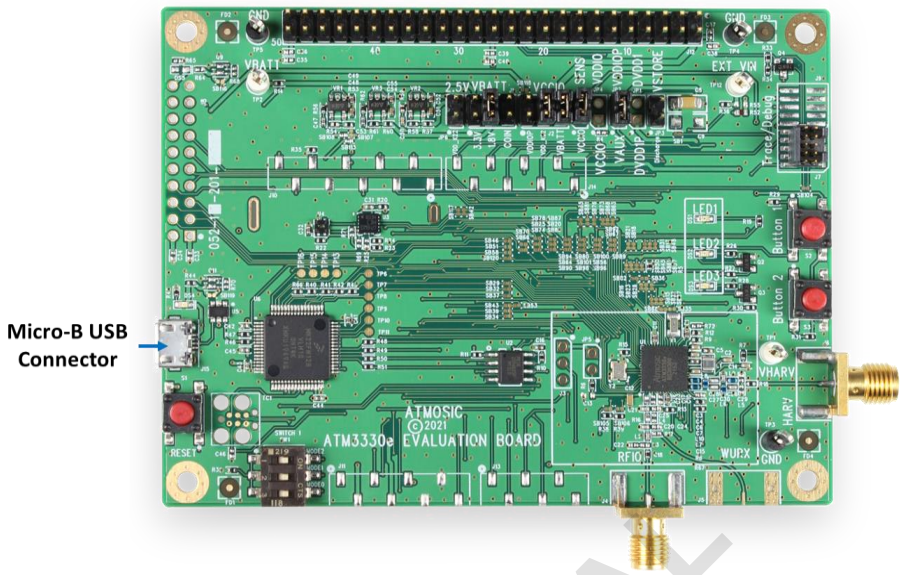


Figure 11 - ATM3330e/ATM3330 Evaluation Board

For Atmosic SDK Windows Installer users, please go to Windows Start -> Atmosic SDK ,run "Setup IDE Environment" and skip step 2-4 below.

2) Install GNU ARM Embedded Toolchain 10.3-2021.07 and add the toolchain path to the user or system path
3) Add a system environment variable: (Replace {*ATM_SDK*} name with the actual installation folder name of Atmosic SDK)
    a) System variable: ATM_SDK_MSYS2
    b) Value: {*ATM_SDK*}\tools\msys2-base-x86_64-{*MSYS2_version*}
4) Edit system environment “Path” to add gcc toolchain in path.
    a) Add “{*ATM_SDK*}\tools\gcc-arm-none-eabi-10.3-2021.07\bin”
5) Reboot PC for system environment variable update.
6) Install Keil MDK-ARM
7) Open Keil project file *example_name*}.uvprojx in {*ATM_SDK*}\platform\atm33\ATM33xx-3\examples\{*example_name*}\[keil_gcc, keil_arm]

3.2 Build and Program Examples

1) Ensure “J-Link driver” is showing in the Windows Device Manager under USB devices.
2) Example project (under keilauto folder)
3) Edit/Compile/Program firmware in Keil IDE

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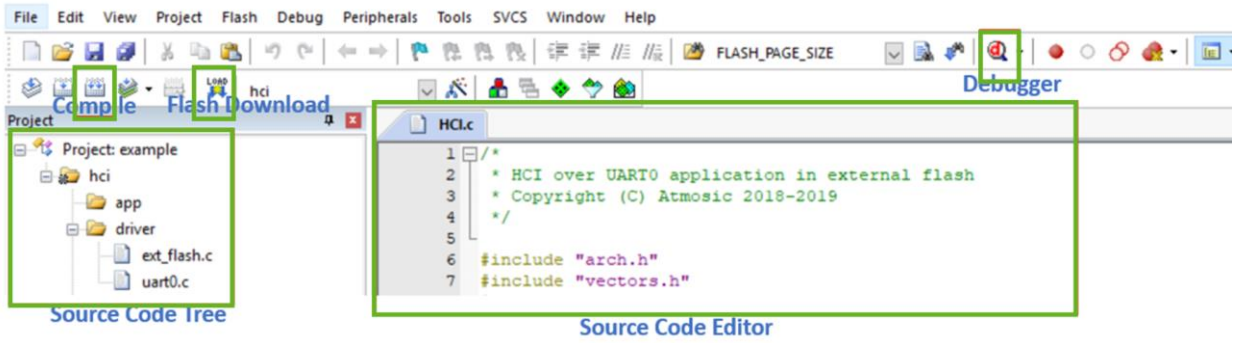
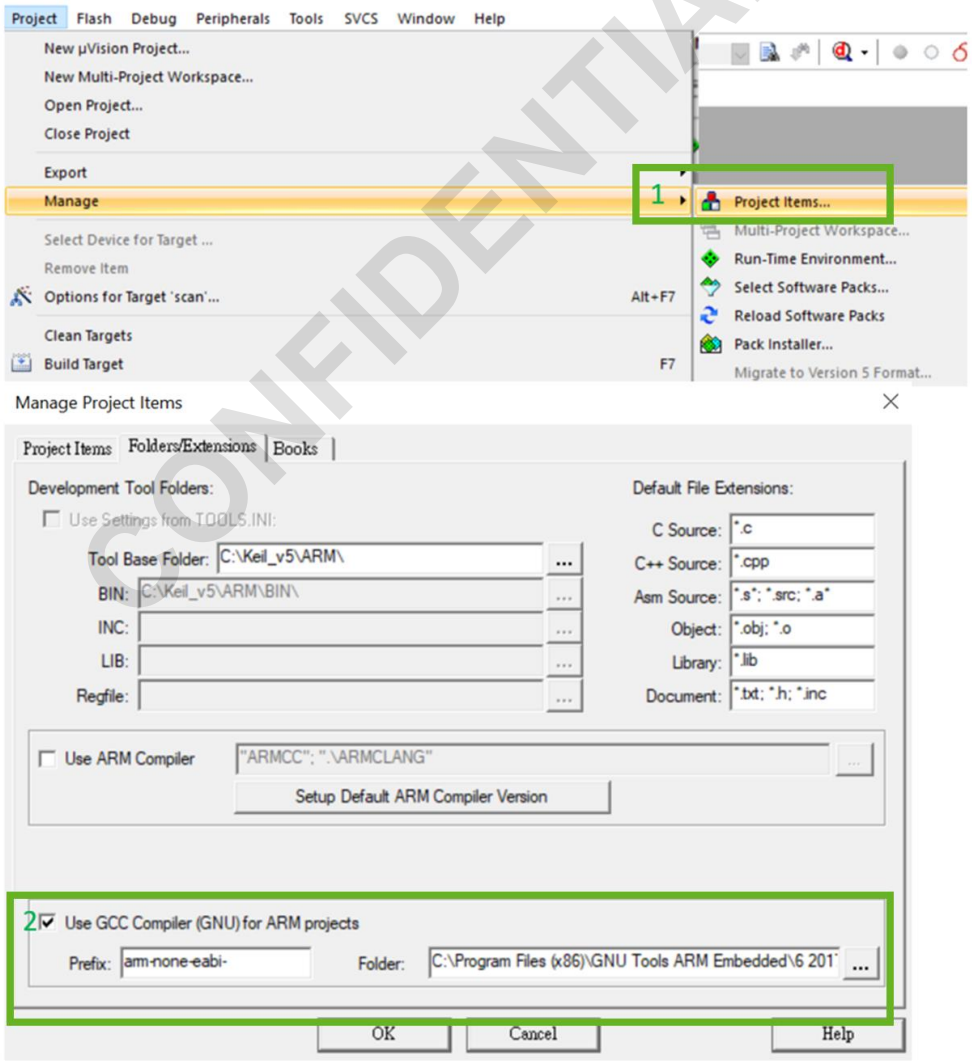


Figure 12 - Keil MDK IDE GUI

3.2.1 Use GCC Compiler in Keil

When creating a new keil project, follow the steps below to select GNU GCC compiler. See Figure 13.



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Figure 13 - Select Compiler

3.3 Release Build

Open the “Option of Target” dialog of Keil IDE

- CC tab:
  - Remove “CFB_DBG” in Preprocessor Symbols Define
  - Add “-flto” option in Misc Controls
- Linker tab:
  - Add “-flto” option in Misc Controls

3.4 Bootloader and NVDS

The corresponding NVDS data (flash_nvds.bin) would be built with the KEIL project generating make command (keil_gcc_gen/keil_arm_gen) and applying the NVDS related make options to the NVDS bin file. The pre-build process would generate the bootloader binary file. In the Keil IDE download process, the bootloader, firmware and NVDS data bin files would be programmed into the device together through J-Link. See Figure 14.

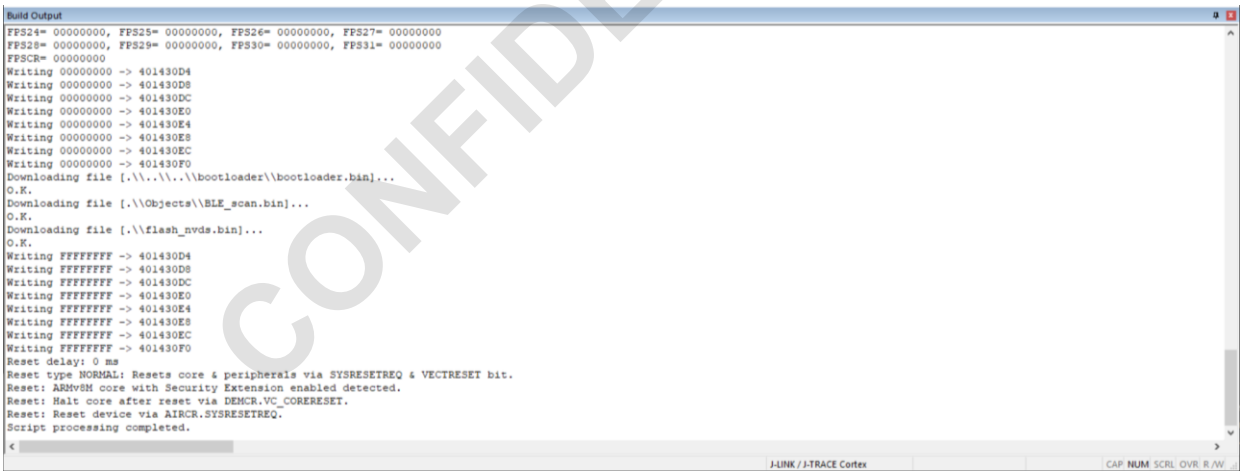


Figure 14 - Program Bootloader and NVDS Data

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Revision History

Date	Version	Description
May 13, 2022	0.60	Initial version created for SDK 5.1.0, which supports ATM2/ATM3 and ATM33 platforms.

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